AFFINE OPENS IN A SEPARATED SCHEME OVER A SEPARATED BASE MEET AFFINELY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4633-4638, following the statement at 4625-4631.

Editorial note. The parenthetical '(and a closed subscheme of U x V)' is supplied by part (1)(b) of the consumed engine card L26387 via the diagonal preimage identification; flagged here.

Editorial note. The step making X -> Spec(Z) separated composes f with S -> Spec(Z) (separated by L26428); the composition step is the engine card L26393 part (1).

Allowed standard prerequisites (editorial).

- Permanence of separation axioms. lemma-separated-permanence (card L26393, cited at ~4634). f separated and S separated (over Spec(Z) by L26428) give X -> Spec(Z) separated.
- Affine intersection characterization of separated morphisms. lemma-characterize-separated (card L26387, cited at ~4636). Part (1) applied to X -> Spec(Z) gives U cap V affine and a closed subscheme of U x V.
- Affine schemes are separated. lemma-affine-separated (card L26428, implicit at ~4637). S separated means S -> Spec(Z) separated; composing gives X -> Spec(Z) separated.

Source reference. Original lemma 01KP: sources/stacks-schemes/pinned-original-source.tex, source lines 4374–4374. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KU: sources/stacks-schemes/pinned-original-source.tex, source lines 4506–4506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism. Assume f is separated and S is a separated scheme. Suppose $U \subset X$ and $V \subset X$ are affine open. Then $U \cap V$ is affine (and a closed subscheme of $U \times V$).*

Proof. In this case X is separated by Lemma 01KU. Hence $U \cap V$ is affine by applying Lemma 01KP to the morphism $X \to \mathrm{Spec}(\mathbf{Z})$. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.